Yanez Multimodal Inorganic Identities Dataset (MIID) Subnet (ت)

Phase 4

V.4.4 – April 2026

Phase 4 — Cycle 4

Continued Adversarial Face Testing: Challenging Bit-Mind Face Deepfake detector and Passive Liveness + **Dual Incentive Mechanism**

At the start of Phase 4 — Cycle 4's execution phase, we carried forward the same core architecture established in prior cycles: validators issue seed face images, miners generate constrained variations, and submissions move through encrypted upload and offline post-validation.

Cycle 4's goal is two fold:

- more data that breaks Bit-Mind's AI detection
- more data that breaks our new passive liveness system.

Heading into Phase 4 — Cycle 3, our primary goals were to increase miner productivity, performance, and overall involvement across the network — and we succeeded. Hot key participation and submission volume both grew substantially, and the positive score rate climbed from 64.01% to 76.09% between Cycle 2 and Cycle 3. Miners weren't just submitting more; they were submitting better, and that combination is exactly the signal-over-noise outcome this phase was designed to produce.

We want to carry that momentum directly into Cycle 4, continuing to produce high-quality adversarial data while making sure miners are set up to succeed in doing it. The challenge ahead is a step harder, but given how strong Cycle 3 turned out, we believe miners are well positioned to keep pushing.

Cycle 4 also marks a structural change beyond the data pipeline: starting July 1, SN54 moves to a dual incentive mechanism. Our Adversarial Data Generation miners keep earning exactly as they do today — nothing about their share changes — while a new, second layer begins rewarding vetted commercial partners for onboarding real, verified humans into the YANEZ proof-of-uniqueness registry. The new layer is funded entirely by burning less, not by taking anything from existing miners. Details below.

Bit-Mind results

On the adversarial side, miners continued pushing against Bit-Mind's deepfake detector, and the results show real progress. Across this cycle's submissions, 6,626 images were classified as non-AI by Bit-Mind — meaning those images successfully passed as real and broke the detector. This confirms that miners are converging on techniques that genuinely stress today's detection systems rather than just producing volume.

What's New in Phase 4 — Cycle 4

Continuous focus on non-AI detection

The main focus heading into this next cycle is non-AI detection itself. In Cycle 3, breaking Bit-Mind's detector earned miners a bonus point on top of their core score — a nice-to-have, not a requirement. That changes now. Going forward, having an image detected as not AI-generated becomes a crucial part of the score itself, not a bonus layered on top of it. This shift reflects where the network's priorities are headed: a face variation that preserves identity and follows the requested transformation is only half the job if it still reads as obviously synthetic to detection systems. Miners should treat detector evasion as a core requirement of every submission this cycle, not an optional stretch goal.

New API for passive liveness

This cycle also introduces a new Passive Liveness (PL) API. The goal of the PL API is to generate a detector capable of identifying whether a submitted face image is a direct capture of a real person or a photo of a screen displaying that face — phone, tablet, laptop, monitor, or TV.

Miner tasks will continue to include screen replay generation, but the objective now goes a step further: miners are being asked to create genuine attack vectors against the passive liveness system itself. This becomes a new UAV category, structured the same way the address UAV worked in earlier phases — miners aren't just generating screen-replay images for their own sake, they're probing for realistic edge cases that the PL detector gets wrong. Specifically, miners will need to produce screen-captured face images that the PL API classifies as not a screen replay. A successful submission in this category functions exactly like an address UAV did under LDS: where it will needed to be manually reviewed afterwards. Full technical detail on the PL API — endpoints, authentication, rate limits, and response format — is available in the API documentation.

What will be asked for structure of rewards

The per-query submission structure stays largely the same. Miners will continue to be asked for the same 5 images:
5 images total:

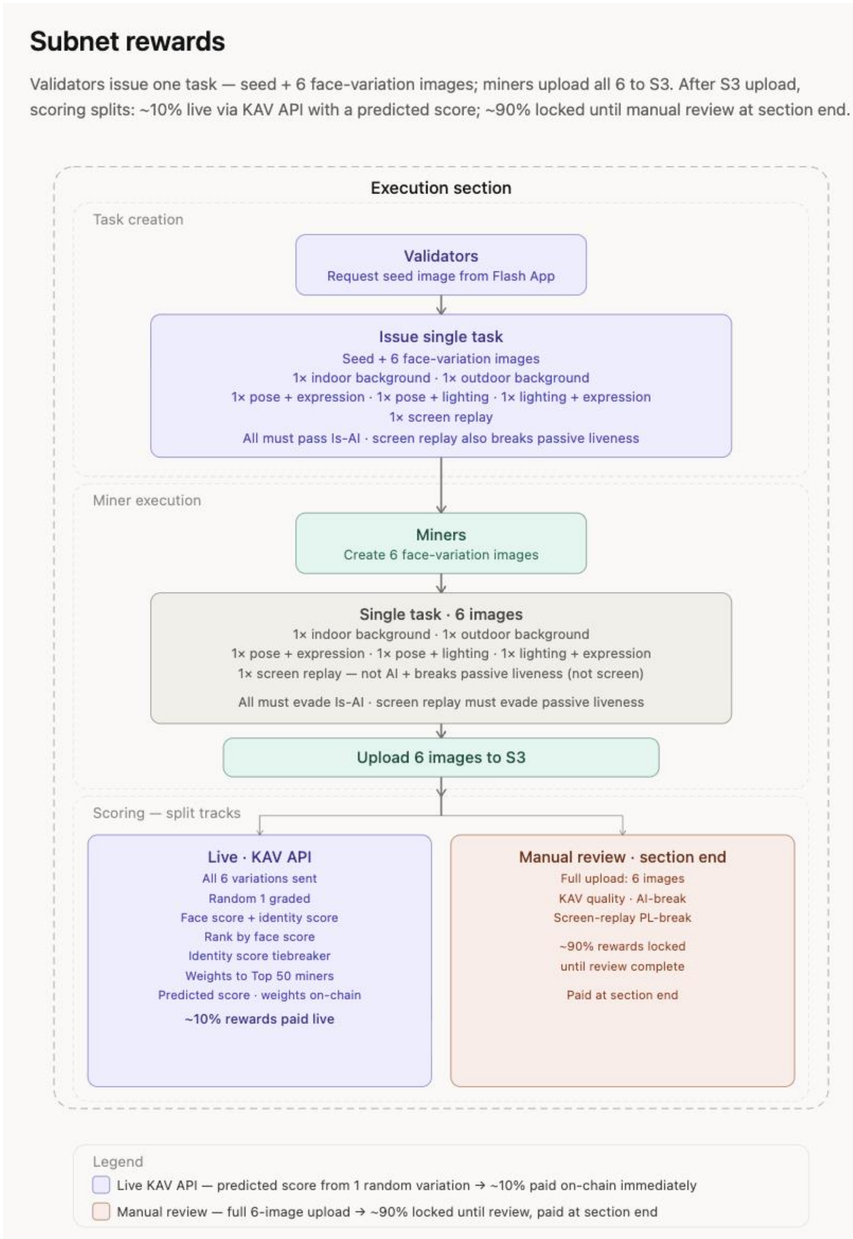
- 2× background edits — detected as not AI
- 1× Pose + Lighting (both together) — detected as not AI
- 1× Pose + Expression (both together) — detected as not AI
- 1× Lighting + Expression (both together) — detected as not AI
- 1× screen replay — not AI **and** breaks passive liveness (not screen)

New this cycle: every single query the screen reply will need to break the passive-livenes — an image specifically intended to be classified as not a screen replay by the PL API, submitted alongside the standard 5.

Structure of rewards KAV → face variations (10% of score)

We are removing the identity-based KAV entirely. Name variations, DOB variations, and address variations — the original Known Attack Vector task — are no longer part of the query. Going forward, KAV is exclusively face variations.

New Structure of rewards: KAV, Face Variation/UAV, New partners



KAV → face variations (10% of score)

The reward structure is changing to match the removal of identity-based KAV. Previously, miners were asked for two separate things: a KAV submission (name, DOB, and address variations), which was scored live and rewarded in that same cycle, and a face variation submission, which was collected but only scored — and rewarded — after manual post-validation in the following cycle. Under that structure, 10% of a miner's reward came from the live KAV result, with the remaining 90% landing later from validated face variation performance.

Under the new structure, KAV itself is the 5 face-variation images. Miners generate these images and submit them to the grading API, which returns a preliminary, automated score. 10% of a miner's reward is paid out based on that live API result — this is now the "KAV" portion in name and function. The remaining 90% is paid out later, after the same images go through full manual post-validation, with an additional bonus applied on top for any submissions that successfully break passive liveness.

Dual Incentive Mechanism — New Partners (Mechanism 2)

This cycle introduces the piece referenced above as “New partners”: starting July 1, SN54 moves to a dual incentive mechanism. Mechanism 1 keeps rewarding our Adversarial Data Generation miners for the identity-testing data they already produce — face variation and liveness — with no change to their share, or to the expected further allocation mechanism. Mechanism 2 is new: vetted commercial partners earn α for every real, unique human they onboard into the YANEZ proof-of-uniqueness registry. The new layer is funded by burning less of the miner emission — not by taking anything from current miners. Of the miner emission, approximately 30% is burned, approximately 35% goes to the data-generation miners, and approximately 35% funds the partner pool — an equal daily α share for both miner types. From July 1, a public, auditable Yanez Incentive Hold (“Hold”) begins accumulating the partner pool so the program has α ready to reward partners from day one and can hold a stable price band per verified human. Daily payouts begin as partners go live (the first two in July), starting through custodied wallets and migrating to fully on-chain rewards.

Where SN54 Stands Today

SN54 is the live name/face identity-testing subnet, and its miners are the core engine — not passive participants, but the ones who make the subnet work and scale. Over recent cycles, they drove a step-change in quality: the leaderboard grew from 8 to roughly 100 miners, predicted data-quality scores rose more than 327%, and miners now produce adversarial face-variation, liveness, screen-capture, and deepfake-pressure data that feeds the systems we build — including a first passive-liveness model from 100,000+ screen-capture samples.
Scoring is not volume-based. Final truth comes from post-validation with a human in the loop: anti-cheat measures, duplication penalties, task adherence, and reputation. We reward the validated signal, not raw activity. Because quality rose, we already lowered burn from 85% to 65%.

The timing is deliberate. As of June 2026, a Bittensor emission change means a subnet that burns miner emission now also gives up chain emission for it. Burning less is therefore doubly rewarded: it funds real, validated work and restores SN54's own emission and price support. SN54 was built in this direction before it became obvious.

The Dual Incentive Mechanism

SN54, starting July 1st, will run two incentive layers in parallel, inside the same subnet emission.

- **Mechanism 1 — Adversarial Data Generation miners (unchanged).** Our KAV and UAV miners keep earning α for identity-testing data — face variation and liveness/screen-capture, plus adversarial and deepfake-pressure data that harden our detection systems. Their share is not diluted.
- **Mechanism 2 — partner referral (new).** Vetted commercial partners earn α for every real, deduplicated human they onboard into the YANEZ PoU/PoH registry, where that human performs billable actions within our partners' products. One verified unique human = one YID. Bots, duplicates, and Sybil farms are filtered out before any α is earned.

The key design move: the new layer is funded by burning less, not by taking a cut from data-generation miners. More partners succeeding makes the pie bigger, not smaller — existing miners keep the share they always had, and the data-generation reward is planned to grow in future cycles as previously specified.

Emission & Burn — the 30 / 35 / 35

Of the subnet's emission (7,200 α/day), miners receive 41% ≈ 2,952 α/day. We split that miner emission three ways — owner and validator emissions are untouched:
Both productive layers receive an equal daily α budget: the Adversarial Data Generation miners ≈ 1,033 α/day and the PoU partner pool ≈ 1,033 α/day (≈ 31,000 α/month). The burn is ≈ 886 α/day. This is funded purely by lowering the burn — data-generation miners stay whole. And post-spec-change, the lower burn also lifts SN54's own chain emission and price support, so the same move that funds partners strengthens the subnet.

The Partner Referral Program

Each YID a partner registers matures for 24 hours, is verified as a unique live human, then is settled once at a single per-YID rate and retired from the payment base (never paid twice). Settlement is daily, and the rate is the same for every partner on a given day:

- **Equal pay, fair by design.** Every partner earns the same α per YID. The best partner wins on volume, never on rate; we lift smaller partners up toward the best rather than capping the best down.

- **A planned per-YID band.** R_min = 0.1 α and R_max = 1.0 α per verified human (R_max will be adjusted later based on demand and utility from our partners). The rate floats with the day's total volume inside that band — the Yanez Incentive Hold (see below) is what maintains the edges.
- **How payouts roll out.** Daily settlement begins as partners integrate (the first two in July). In the first phase, settlements are disbursed to partner coldkey (CK) public and auditable wallets. As the program scales, we migrate everyone to fully on-chain rewards, where partner UIDs receive weight-based emissions directly.

Why We Start With Vetted, Committed Partners

Biometric onboarding has to be earned by utility, never bought. We do not — and will not — pay individuals to hand over their biometrics. Programs that paid people directly to enroll their faces or irises (or other biometric signals) have drawn regulatory enforcement in multiple jurisdictions, where the monetary incentive was found to undermine the validity of consent. Our reward flows to partners for bringing real, consenting users into a genuine service — never to users for submitting biometrics.

That is why we launch with vetted, committed partners. Each partner already runs a product its users want, and the YANEZ PoU/PoH is used from day one inside that product — to verify a real human, sign in, or prove uniqueness. The biometric step exists because the user is getting a service, which keeps consent freely given and the purpose legitimate — and it means the α has real demand immediately: the users being onboarded are the users transacting.

- **Utility from day one.** Partners use the PoU/PoH for their own users the moment they integrate — real verification volume, not a stockpile of unused enrollments.
- **Consent that holds up.** Enrollment is tied to a service the user chose, not to a payment — the design regulators look for, and the trap others have fallen into.
- **Purpose, not harvesting.** Vetted partners bring real, active, deduplicated humans; collection is grounded in genuine use, not random data harvesting.

Principle: we reward partners for utility — bringing users into a real service — never users for giving biometrics.

Privacy-Preserving Proof-of-Uniqueness

Every signup is biometrically verified as a unique, live human before it counts. One YID = one human; duplicates, bots, and Sybil farms are removed before any α is earned. The check is privacy-preserving: the registry holds a protected, searchable uniqueness representation and an opaque per-human identifier (YID) — not raw biometrics. Biometric processing happens on the user's device; the network learns only "is this a new, unique human?", not who they are. This is what gives the α a hard, non-gameable meaning — each unit of partner reward corresponds to a real, distinct person.

The Yanez Incentive Hold (“Hold”) — an Emission-Funded Buffer

Partners don’t bring cold sign-ups; they bring their existing active users — real, already-engaged people who use the verification from day one. When a partner switches on, many users arrive at once, and each user we reward costs α. The Hold is what guarantees that we have α ready to incentivize our partners’ customer base use of the service. Crucially, it is funded by emissions and nothing else — the daily pool and the surplus it banks. We never feed it α from any outside source. It plays three coupled roles.

Role 1 — the high-volume buffer. From July 1, the pool accrues to the Hold before partners switch on. When a partner brings a large group of active users on day one, the Hold already holds the α to reward them immediately. Month-1 accrual is ≈ 31,000 α — the buffer that lets payouts start strong instead of waiting for the pool to fill in real time.

Role 2 — holding the planned per-YID band. The Hold is exactly what lets us control the min and max per YID as planned. On light days, the rate would exceed R_max, so we cap at 1.0 α and bank the surplus (Hold IN). On heavy days a surge would push the rate below R_min, so the Hold tops up to hold the 0.1 α floor (Hold OUT). Every day, the books close exactly — nothing leaks or is minted.

Role 3 — buffer when onboarding is quiet, fuel when it surges. In quiet periods (partners bring few users), the pool isn’t wasted — it banks; in surge periods, that banked α funds the floor. The Hold smooths month-to-month swings so the program remains durable and the value is retained rather than lost.

There is a sustainable ceiling — about 10,330 verified humans/day at these parameters — above which even a full Hold cannot hold the floor indefinitely; metered partner release schedules keep volume near capacity, where the prevailing rate is healthiest.

Emission-funded and auditable: the Hold has exactly one source — surplus from the program’s own emission pool — and the floor it pays is emission too. No owner emission, no fresh minting, no outside α is ever fed in. Its address is public from launch, so anyone can audit what is accumulated and what is paid.

The Closed Loop — Earn, Spend, Settle

This is more than a referral scheme because partners also use the system. A partner earns α by bringing unique humans in; the partner then spends the earned α on usage: every verification, every signing transaction, every proof-of-uniqueness call consumes α.

When a partner’s users transact, the consumed α is moved from the partner’s wallet into the Yanez PoU revenue wallet — YANEZ earns on the transaction itself. If usage outgrows what the partner earned from registrations, the partner settles the difference in α or USD. That settlement is handled as YANEZ revenue, with a defined share (≈ 20%) bought back into the subnet.

So the demand side is real: actual operators with day-to-day verification volume that has to settle — not speculators holding for upside. The token has daily utility, not reward sitting idle in a wallet.

Transparency, Wallets & Roadmap

- **Wallets, never co-mingled.** Wallet A — the Yanez Incentive Hold (emission-funded buffer; public, auditable). Wallet B — per-partner (earned α plus any Hold top-ups; α is spent on usage, never sold). The Yanez PoU revenue wallet — where α consumed by usage and partner settlements (α/USD) land. Emission and revenue stay separate.
- **Public auditability.** The Hold address is published at launch. As each partner integrates, its hotkey/coldkey is published too, mapped to the contracted cycle and the actual PoU/PoH usage it represents.

Roadmap:

- **July 1, 2026 —** Hold begins accumulating the PoU pool (Month 1 = accrue, settle at the month close).
- **Mid- and end-July —** First two commercial partners integrate; daily settlement begins via coldkey (CK) wallets.
- **Next —** Migrate all partner rewards fully on-chain (weight-based emission to partner UIDs); publish wallets.
- **Later —** We're evaluating several approaches for permissionless miners to help grow the network by bringing users to a utility/service app — not by harvesting biometrics for reward; details in the coming weeks. Data-generation rewards are also planned to grow.

Timeline

June 6: Phase 4 Cycle 3 intake stops (post-validation/leaderboard/reporting continue).

June 6 → July 1: Phase 4 Cycle 4 Sandbox (calibration + stability testing). The normal KAV online flow remains unchanged, and the Burn ratio will be set to 65%.

July 1 – Aug 1: Phase 4 Cycle 4 execution goes live; reward allocation begins using validated Phase 4 Cycle 3 Face Variations reputation (alongside the ongoing online flow). The burn ratio will stay 65%.

Phase 4 — Cycle 3

Continued Adversarial Face Testing: Deep-fake detection

We recently closed Phase 4 — Cycle 2 and are now live in Phase 4 — Cycle 3. Cycle 3 keeps the same core architecture, miner workflow, and post-validation pipeline introduced in earlier Phase 4 cycles — validators issue seed face images, miners generate constrained variations, and encrypted submissions are evaluated through offline validation. What changes in Cycle 3 is the focus: this cycle is about the miner experience — making it easier to start, easier to iterate, and easier to see how your submissions are performing in real time.

Another important part of Cycle 3 is deep-fake detection pressure-testing. We send synthetically generated images to the BitMind deepfake detector, the goal is to find where the detector fails, then use those failures to make the system stronger. In a world where "seeing is believing" is no longer enough, this is a crucial step for building safer and more reliable identity verification.

As with previous cycles, the goal remains signal over noise: high-quality, constraint-adherent variations that expose realistic failure modes in biometric verification systems.

What changes now (Phase 4 Cycle 3)

- **Phase 4 Cycle 2 intake stops:** April 20 (post-validation, leaderboard, and reporting continue).
- **Phase 4 Cycle 3 Sandbox: April 20 → May 8** (calibration + stability testing).
- **New Miner dashboard:** Live dashboard for miners face variations giving them a estimate score
- **New BitMind Deep-fake detector:** Miners will get a bonus point if there images are detected as not ai
- **Cycle 2 transformation families: pose_edit, lighting_edit, expression_edit, background_edit** with **Light / Medium / Far** intensity bins. Also **screen_replay** with **phone / tablet / laptop / monitor / TV** devices

What's New in Phase 4 — Cycle 3

Cycle 3 is focused on **YOU — the miners**. We saw that miner participation, while growing, is still lower than we want, so this cycle is built around three goals:

- Make it **easier to get started**.
- Help existing miners **perform better**.
- Bring **more miners into the network**.

Recent base-code improvements were just the beginning — more is coming throughout the cycle.

Better Miner Experience BitMind results:

Setup and iteration flows are being simplified so miners can spend their time on output quality rather than on friction. Expect ongoing updates to the miner reference implementation and onboarding tooling.

Guides, Resources & Videos

We will be dropping step-by-step guides and reference materials throughout Cycle 3 so new miners can ramp quickly and existing miners have a clear reference for best practices and avoiding common pitfalls.

Live Performance Dashboard

A real-time dashboard is coming so miners can:

- Track performance live across queries.
- See how individual variations are scoring.
- Understand what is working — and what is not.

This turns scoring into a continuous feedback channel during the cycle, not just at the end.

Multi-Detector Evaluation

Every submission in Cycle 3 will be evaluated against multiple AI detectors, including:

- **Face-specific models**
- **General AI-detection systems**

Per-detector results will be visible on the live dashboard, giving miners direct insight into which detectors their variations break and which ones still catch them.

Bonus Incentive: Challenge BitMind

+1 bonus point for any variation that breaks BitMind's deepfake detector. Miners who push the limits get rewarded directly — this is a deliberate signal that high-effort, adversarial work is what we want to concentrate emissions around.

BitMind can label a face variation as "real" even when the image is generated. That label is not provenance — it does not mean “this came from a camera.” The validator seed is an identity anchor; the miner’s job is to preserve identity while making the output look natural. A “real” read can still be useful: it means the variation is realistic enough to stress today’s AI-image detectors. We treat this as controlled adversarial / detector stress-testing, not a final truth about where the pixels came from.

BitMind are live feedback and bonus signals. Final scoring stays on YANEZ post-validation — identity preservation, requested transformation adherence, quality, reputation, and anti-cheat — same philosophy as prior cycles.

Cycle Growth at a Glance

Phase 4 has grown sharply cycle over cycle. Summary of leaderboard participation:

Metric	Phase 4 — Cycle 1	Phase 4 — Cycle 2
Miners on the leaderboard	8	**100**

Cycle 2 was the first time the network crossed 100 miners on the leaderboard. Cycle 3’s focus on onboarding, tooling, and live feedback is explicitly designed to keep that growth curve moving — both by retaining returning miners and by lowering the barrier for new ones.

Weight Shift: KAV vs Face Variation/UAV

We also updated the score weighting to put more emphasis on adversarial image performance:

- KAV weight: 0.15 → 0.10
- Face Variation/UAV weight: 0.85 → 0.90

Previous formula:
$\text{Miner\_rep} = 0.85 \cdot \Delta\text{face\_score} + 0.15 \cdot \Delta\text{rep\_KAV}$

New formula:
$\text{Miner\_rep} = 0.90 \cdot \Delta\text{face\_score} + 0.10 \cdot \Delta\text{rep\_KAV}$
This change reflects Phase 4 priorities: stronger focus on Face Variation and UAV signal, and less emphasis on baseline KAV contribution

Timeline

April 20: Phase 4 Cycle 2 intake stops (post-validation/leaderboard/reporting continue).

April 20 → May 6: Phase 4 Cycle 3 Sandbox (calibration + stability testing). The normal KAV online flow remains unchanged, and the Burn ratio will be set to 65%.

May 6 – June 6: Phase 4 Cycle 3 execution goes live; reward allocation begins using validated Phase 4 Cycle 2 Face Variations reputation (alongside the ongoing online flow). The burn ratio will stay 65%.

Phase 4 — Cycle 2

Continued Adversarial Face Testing: Screen Replay

Phase 4 — Cycle 2 continues the adversarial biometric testing framework introduced in Cycle 1. The core system architecture, miner workflow, and post-validation pipeline remain unchanged. Validators will continue issuing seed face images, miners will generate constrained variations, and encrypted submissions will be evaluated through offline validation.

Cycle 2 focuses on expanding the realism of attack simulations rather than introducing entirely new task structures. Specifically, Cycle 2 introduces a new variation family designed to simulate screen replay attacks, one of the most common real-world bypass attempts in modern identity verification systems. This change expands the adversarial coverage of Phase 4 while maintaining the same constraint-driven evaluation framework used in Cycle 1.

As with previous cycles, the goal remains signal over noise: high-quality, constraint-adherent variations that expose realistic failure modes in biometric verification systems.

What changes now (Phase 4 Cycle 2)

- **Phase 4 Cycle 1 intake stops:** March 2 (post-validation, leaderboard, and reporting continue).
- **Phase 4 Cycle 2 Sandbox: Mar 2 → Mar 16** (calibration + stability testing).
- **New face variation:** new variation designed to simulate realistic screen-captured faces.
- **Cycle 2 transformation families: pose_edit, lighting_edit, expression_edit, background_edit** with **Light / Medium / Far** intensity bins. Also **screen_replay** with **phone / tablet / laptop / monitor / TV** devices
- **Location track STOPS:** No more miners submit **UAVs addresses**
- **Rewards signal evolution:** Phase 4 Cycle 2 execution will begin incorporating **validated UAV quality from Phase 4 Cycle 1** into reward/reputation allocation (thresholds/metrics will be disclosed later to reduce gaming).

New Face Variation: screen_replay

We are introducing a new variation designed to simulate realistic screen-captured faces. This variation represents scenarios where a face image is displayed on a physical device and then captured again by a camera. These types of captures commonly appear in real-world replay attacks and display recordings.

Overview

The generated image must appear as if the face is being displayed on a **real device** and then photographed or recorded.

Examples of valid devices include:

- phone
- tablet
- laptop
- monitor
- television

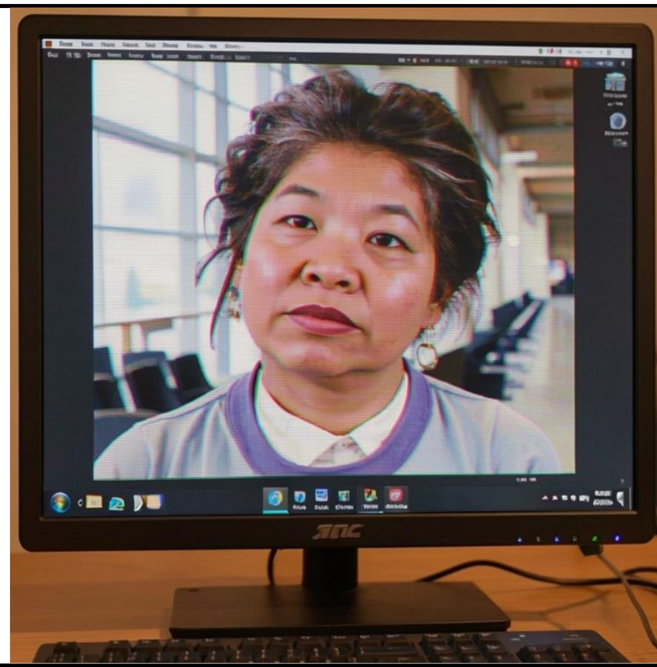
The face must remain **clearly visible and large enough for reliable detection**, and the **identity must remain strongly matchable to the original seed**.

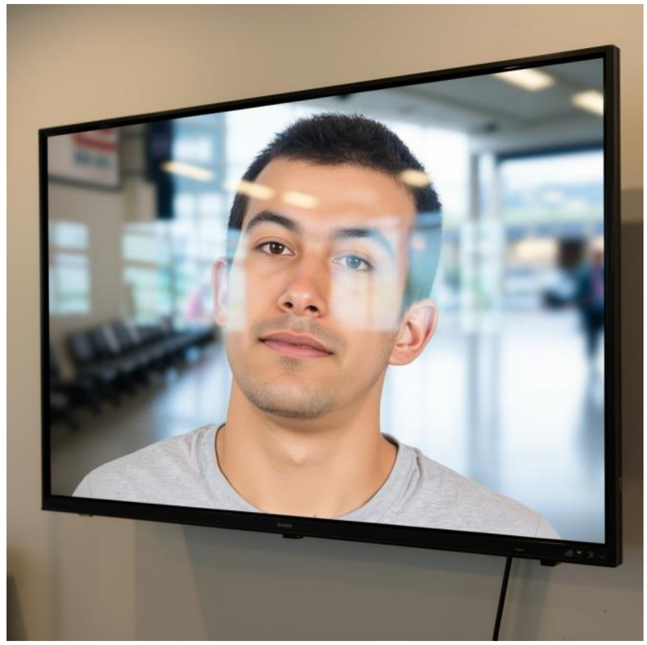
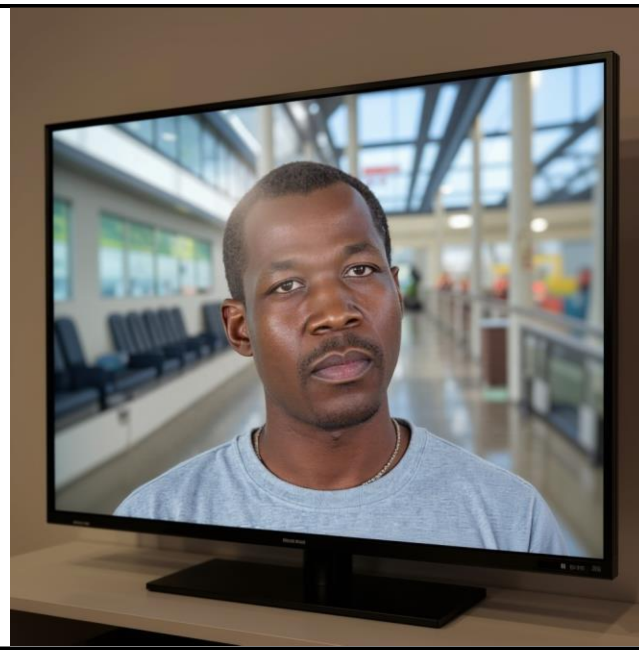
screen_replay Requirements

Requirement	Specification
Variation type	screen_replay
Required metadata	device_type: phone | tablet | laptop | monitor | tv
Visual cue requirement	Must exhibit **at least 2** realistic cues
Face constraint	Face must remain the dominant object
Matchability constraint	Identity must remain matchable to seed

Visual Cue Requirement (≥2 Required)

To simulate a realistic display capture, each image must visibly contain **at least two** of the following cues:

Cue	Description	
Moiré / pixel grid	Interference patterns created when camera sensors capture screen subpixels	

Screen glare hotspots	Specular reflections or bright glare areas on the display surface	
Perspective / keystone distortion	Geometric distortion caused by filming the display at an angle	

Gamma / contrast shift	Brightness, contrast, or color differences typical of display capture	
Edge / crop cues	Device borders, bezels, or screen framing visible in the image	

UAV for Address is not required anymore

During previous phases we accepted UAV address submissions as part of the broader evaluation process. After extended testing and analysis, we did not observe consistent, high-quality examples capable of reliably breaking the current Location Detection System

(LDS). While some isolated cases were submitted, they did not demonstrate a repeatable or scalable method capable of bypassing the system under realistic conditions.

As a result: **UAV address submissions will no longer be accepted, moving forward.**

This vector will be removed from the current reward structure while we continue to focus development on the face variation generation tasks

Additionally, after completing the grading process we found that address-related scoring performance was extremely low across the board. All participants either received negative scores in the address category or did not participate in that portion at all. Because of this, the influence of the location component on overall scoring will be reduced from 20% to 5%.

The scoring formula has therefore been updated as follows:

Previous formula:

$$\Delta rep_{p4} = 0.80 \cdot \Delta rep\_face + 0.20 \cdot \Delta rep\_loc$$

New formula:

$$\Delta rep_{p4} = 0.95 \cdot \Delta rep\_face + 0.05 \cdot \Delta rep\_loc$$

This adjustment reflects the current performance distribution and places greater emphasis on the face variation tasks, which remain the primary focus of the evaluation.

Timeline

March 2 : Phase 4 Cycle 1 intake stops (post-validation/leaderboard/reporting continue).

Mar 2 → Mar 17: Phase 4 Cycle 2 Sandbox (calibration + stability testing). The normal KAV online flow remains unchanged, and the Burn ratio will be set to 70%.

Mar 17 – April 20: Phase 4 Cycle 2 execution goes live; reward allocation begins using validated Phase 4 Cycle 1 UAV reputation (alongside the ongoing online flow). The burn ratio will be lowered to 65%.

Phase 4 — Deepfake / Face-Based Adversarial Testing for KYC: Synthetic Biometric Variation Generation

Phase 4 expands MIID from demographic, text-only identity attributes (name/DOB/address) to the hardest KYC entry point: the face. Based on feedback from pilot customers/design partners, we are prioritizing synthetic adversarial testing across identity—not "more data," but data that breaks real IDV systems in the same ways attackers do.

Deepfakes erode trust because they compress "high-effort fraud" into repeatable pipelines. MIID Phase 4 responds by introducing structured face-variation generation with identity preservation: miners receive validator-provided seed face images and generate constrained variations across a small set of transformation families (starting with pose, lighting, expression, and background), each requested at Light / Medium / Far intensity. These families are designed to simulate both high-fidelity re-synthesis and low-resource/noisy attack conditions—while keeping evaluation grounded in constraint adherence and post-validation.

As in Phase 3, the goal remains signal over noise. Phase 4 is designed so that "high effort" work actually pays in execution: rewards and reputation will increasingly concentrate around miners who deliver rare, high-impact submissions—constraint-following face variations and UAVs unknown to LDS V1—rather than volume, recycling, or exploit loops. Phase 4 also continues the location track in parallel, so the subnet advances toward a unified identity-testing loop across biometrics, location, and identity attributes.

What changes now (Phase 4 Cycle 1)

- **Phase 3 Cycle 1 intake stops:** Dec 31 (post-validation, leaderboard, and reporting continue).
- **Freeze window:** Dec 31 → Jan 8 (no new Phase 3 UAV intake; validation continues).
- **Phase 4 Cycle 1 Sandbox: Jan 8 → Jan 22**(calibration + stability testing).
- **New modality (face):** validators provide **seed face images**; miners generate requested face variations under constraint.
- **Cycle 1 transformation families: pose_edit, lighting_edit, expression_edit, background_edit** with **Light / Medium / Far** intensity bins.
- **Location track continues:** miners submit **UAVs unknown to LDS V1** (higher bar; fewer, higher-impact discoveries).
- **Rewards signal evolution:** Phase 4 execution will begin incorporating **validated UAV quality from Phase 3 Cycle 1** into reward/reputation allocation (thresholds/metrics will be disclosed later to reduce gaming)

Motivation and Threat Model (Why Phase 4 Starts With the Face)

Phase 4 is driven by a simple observation from pilot customers/design partners: the face is now the highest-risk KYC entry point, because modern deepfake pipelines can turn "high-effort fraud" into repeatable, scalable production. In other words, "seeing is believing" is no longer a safe assumption—and resilient KYC requires testing systems against the same operational realities attackers use.

In our public framing, we highlighted the scale of deepfake-driven losses (e.g., "$1.56B lost to deepfake attacks in 2025 alone"), not as a marketing statistic, but as a signal that the attack surface has already moved.

What We Are Testing

MIID Phase 4 targets **face-based adversarial testing under identity preservation constraints**. The goal is not to generate "pretty images," but to generate **structured, labeled synthetic face images** that expose failure modes in real identity verification stacks (liveness checks, face match, capture conditions, pre-processing, downstream decision thresholds). This expands MIID's adversarial loop from text-only identity attributes to **multimodal identity integrity**.

Attacker Model: Practical Deepfake Pressure, Not Lab Demos

We assume adversaries can use:

- **High-fidelity re-synthesis** (strong generators, curated prompts/pipelines, tuned outputs)
- **Low-resource / noisy fakes** (compressed, recaptured, poor lighting, weak phones, messy backgrounds)
- Rapid iteration to find the "cheap path" through an IDV system's weakest link

Phase 4 therefore starts with a constrained set of transformation families (pose, lighting, expression, background) because these cover a large portion of real-world variance and adversarial "pressure points" *without* immediately escalating to more complex families (swap/morph/recapture), which are reserved for later cycles.

Why Synthetic Adversarial Data (and Why "Signal Over Noise" Still Rules)

Static datasets and one-off benchmarks fail for the same reason most defenses fail: **attackers adapt faster than evaluation**.

MIID's approach is to make evaluation **cycle-based and adversarial by construction**, so the subnet learns from what breaks systems, not from what is easiest to mass-produce.

That is also why Phase 4 keeps the Phase 3 philosophy intact: rewards and reputation must focus on **rare, high-impact signals**, not on volume, duplicates, recycling, or exploit loops.

Economic Reality: Emissions Dilution and Subnet Health

This work has an economic constraint: under **dynamic TAO (dTAO)**, subnet rewards and participation dynamics are coupled to **alpha token flows and reserve pool mechanics**; excessive low-signal extraction can contribute to volatility, dilution concerns, and weaker long-term emissions dynamics into the pool.

Phase 4 explicitly aligns incentives so that **high-effort, high-signal adversarial work becomes the economic equilibrium** as the reputation and post-validation layers mature—reducing the "race to the bottom" that noise-heavy tasking creates.

Success Criteria (High-Level, Non-Gameable)

Because thresholds can be gamed, Phase 4 describes acceptance in **constraint-based terms** rather than publishing hard lines. At a high level, post-validation evaluates:

- **Identity preservation** (embedding-consistency style checks, without exposing exact thresholds)
- **Requested transformation adherence** (did the miner generate the right type/intensity family)
- **Quality + anti-cheat** (detecting duplicates, recycling patterns, and other exploit behaviors)

Phase 4 System Architecture and Data Flow

Phase 4 adds a **face-variation channel** to MIID while preserving the subnet's core operating model from prior phases: **validators stay lightweight and fast**, and **ground truth is established later via post-validation**. The result is a pipeline that can run at subnet speed without turning validators into heavy image-processing nodes.

At a high level, Phase 4 introduces a new "commit → store → validate" loop:

- **Commit (online, fast):** validators issue constrained requests; miners generate and commit outputs using integrity primitives; validators only verify submission integrity and continue normal online scoring behavior.

- **Validate (offline, authoritative):** YANEZ post-validation decrypts, verifies, evaluates constraint adherence + anti-cheat, and updates reputation for the next cycle.

Face-variation: End-to-End Flow (Validator → Miner → Storage → Post-Validation)

1. **Challenge packaging (Validator)**: Validator selects a **seed face image** and sends a request containing:
   - Seed face image (the "identity anchor")
   - A **dynamic subset** of requested variation families (2–4 from pose/lighting/expression/background)
   - **Intensity bins** per requested family (Light / Medium / Far)
   - Challenge identifiers / commit-time fields needed by the cycle (e.g., reveal window parameters where applicable)
2. **Variation generation (Miner):** Miner generates the requested set of images **locally**, with two primary constraints:
   - Preserve identity (seed subject remains recognizable)
   - Follow the requested transformation type + intensity intent
3. **Commit + store (Miner → S3):** Instead of returning images directly over the synapse, miners:
   - Encrypt outputs
   - Upload encrypted artifacts to storage
   - Return **only** storage references and integrity metadata to the validator
4. **Online path (Validator: lightweight validation + operation continuity):** Validators do *not* download, decrypt, or evaluate images in real time in the first few cycles:
   - Ensure the miner returned the expected references/metadata
   - Continue normal online operation (including the existing KAV workflow)
5. **Offline Validation (YANEZ post-validation):** After the reveal/collection window:
   - Download encrypted artifacts from storage
   - Decrypt (batch)
   - Verify integrity and ownership
   - Evaluate constraint adherence + anti-cheat signals
   - Produce post-validation outcomes used to update reputation for subsequent execution cycles

Online vs Offline Responsibilities

Phase 4 makes the boundary explicit:

- **Online (Validator):** operates the subnet; keeps checks on structural and non-gameable; avoids heavy computation and avoids revealing other miners' work during the submission window.

- **Offline (YANEZ post-validation):** establishes truth; performs image-based evaluation and anti-cheat at batch scale; updates reputation for the next cycle.

This separation is a design choice to prevent a "race to exploit whatever validators can score instantly" and to ensure incentives converge toward an adversarial signal that survives offline review.

Integrity and Fairness Primitives

Phase 4 relies on a set of primitives that keep the pipeline secure and economically fair without over-specifying thresholds:

- **Reference-based submission (S3 pointers, not image payloads):** keeps validators fast and makes batch truthing feasible.
- **Ownership + integrity binding (wallet signing + hashes):** binds each stored artifact to the miner identity and prevents tampering/re-attribution.
- **Delayed reveal via encryption:** protects miner work during submission, reduces copy/relay behaviors, and supports fair "all-at-once" batch comparison during post-validation.

Acceptance is expressed as constraint adherence rather than fixed published thresholds. Thresholds/metrics can be disclosed later once the system is stable and harder to game.

Protocol and Interface Changes (Phase 4 Cycle 1)

Phase 4 extends the existing **IdentitySynapse** contract to support a **face-variation channel**, while keeping the **Phase 3 identity-attribute path (KAV + UAV)** intact. Validators remain lightweight: **they do not download/decrypt/score images online**; they only verify structural integrity and continue normal online scoring.

IdentitySynapse extensions (request → response contract)

New/extended request fields (additive):

- challenge_id (cycle-scoped identifier)
- seed_identity (existing: name/DOB/address seed context continues unchanged)

- seed_face , the "identity anchor", provided by validator inline (e.g., base64 payload) in the identity synapse.
- variation_requests[]:
    - family ∈ {pose_edit, lighting_edit, expression_edit, background_edit}
    - intensity_bins ⊥ {Light, Medium, Far} (requested bins per family)
    - validator selects a **dynamic subset** (2–4 families per challenge)
- drand_target_round (used for timelock encryption / delayed reveal

New/extended response fields (additive):

- kav_variations (unchanged)
- uav_submissions (unchanged; location track continues)
- s3_submissions[] (instead of raw image payloads), where each entry includes:
    - s3_key (encrypted artifact location)
    - image_hash
    - signature (miner wallet signature over commit fields)
    - path_signature (binds the submission path to the miner + challenge to prevent hijacking)
    - optional: family, intensity, mime, size (for validation bookkeeping)

This matches the Phase 4 "store by reference" design: encrypted images land under a challenge_id / miner_hotkey / path_signature structure, and validators only receive references + integrity metadata.

Track A — Face Variation Generation task spec

What miners do (new modality):

- Input: a validator-provided **seed face** (identity anchor).
- Output: constrained variations across the requested families and intensity bins:
    - Families (Cycle 1): pose_edit, lighting_edit, expression_edit, background_edit
    - Intensity bins: Light / Medium / Far
- Constraints (expressed non-gameably):
    - **Identity preservation** (seed subject remains the same person)
    - **Transformation adherence** (match requested family + intended intensity)
- Submission mechanics:
    - hash + sign → **encrypt with drand timelock** → upload encrypted artifacts to S3 → return only refs + commit metadata.

[Face UAV Score is binary (PASS/FAIL)](#) *and is determined by two checks:*

1. **Identity preservation (ArcFace match)**
    - Compute face similarity using **ArcFace** (via the **InsightFace** implementation: deepinsight/insightface).
    - **PASS** if similarity ≥ **0.70** (default threshold).

Max-intensity exception: for certain **MAX intensity** labels where the transformation can legitimately reduce similarity (e.g., extreme pose / lighting / expression), the acceptance threshold may be relaxed to **0.50–0.60** for that request.

1. **Adherence to the requested label**
    - The generated image must match the **requested transformation label** (e.g., pose_edit, lighting_edit, expression_edit, background_edit) and the intended **intensity level** (Light/Medium/Far/Max as applicable).
    - **PASS** only if the label/adherence is correct for the request.

Final Score (per sample):

While **face acceptance is binary (PASS/FAIL)**, Phase 4 also applies a **post-validation "validation_score"** to support miner reputation updates (exactly like Phase 3 UAV).

Face validation_score (manual + post-validation):

- **+5 (Perfect):** ArcFace match passes the applicable threshold **AND** the sample adheres to the requested label/intensity.
- **+3 (Good / partial credit):** near-miss cases where **identity match OR adherence** is not fully met (e.g., correct identity but adherence is a little bit off, or correct label but identity similarity falls slightly below the target threshold for that request).
- **−5 (Cheat / fake / noise):** wrong identity, spam/noise payloads, obvious garbage, or exploit patterns. **Duplicates/replays** are treated as penalties consistent with Phase 3 policy.

Δrep mapping (same table as Phase 3 UAV):

- **+5 → +1.00**
- **+3 / +4 → +0.40**
- **−5 → −0.50**

Manual post-validation:

As in **Phase 3**, all submissions will also undergo **manual validation** to confirm correctness, catch edge cases, and enforce anti-cheat expectations—especially around **max-intensity** requests where similarity thresholds may be conditionally relaxed.

Track B — LDS v1 “unknown UAVs” continuation (higher bar)

Phase 4 continues the location/UAV learning loop in parallel, but now against an updated baseline: **LDS v1 (beta)** is released after Phase 3 Cycle 1 post-validation, seeded by validated UAVs and the growing known_uav_corpus, so “unknown” means unknown to **LDS v1**, not v0.

In Cycle 1, validators will use LDS v1 as the baseline address-validation API (for KAV address checks and for offline UAV verification). Miners will also have **limited, rate-limited access** to the same LDS v1 API so they can test hypotheses and then submit UAVs designed to attack it—i.e., realistic edge cases that are **unknown to LDS v1**, not recycled patterns.

UAV submission definition (unchanged, carried forward):

- A look-alike/variant of a real address that appears legitimate but fails API validation
- A short explanation label (why it could be real)
- Longitude + latitude of the source address used to create the variant

We keep the **post-validation** principle so execution can’t be gamed mid-cycle, and so validated UAVs can be consistently absorbed into the next LDS iteration.

<u>**Location/address UAV scoring in Phase 4 follows the same Phase 3 post-validation framework.**</u>

Submissions are processed through the following grading flow: **(1) duplication checks**, **(2) region/coordinate validation**, **(3) semantic spam filtering**, then **(4) manual validation & scoring**, where each UAV is assigned a **validation score from −5 to +5**.

As in Phase 3, this process remains **heavily manual** for edge cases and exploit detection, and is designed to preserve **signal over noise** while discouraging spam, duplicates, invalid coordinates, and recycled patterns.

Validation outcomes then map into **reputation deltas (Δrep)** (e.g., **+5 → +1.00**, **−3 duplicate → −0.10**, **−5 cheat/fake → −0.50**), which roll up into a miner’s **rep_score** and **tier** for reward weighting—consistent with the Phase 3 reputation-based reward system.

Phase 4 Unified UAV Reputation Signal ~~(80% Face / 20% Location)~~ (95% Face / 5% Location) ** (Updated March 14, 2026)

In Phase 4, the final UAV reputation signal used for execution-stage emissions is computed as a weighted blend of both tracks: Face and Location/Address.

Specifically, after manual post-validation (as in Phase 3), each miner receives a Face validation outcome (e.g., +5 / +3 / −5) derived from ArcFace identity preservation + label adherence, and a Location/Address UAV outcome scored using the unchanged Phase 3 UAV framework (also mapped into the same Δrep mechanics).

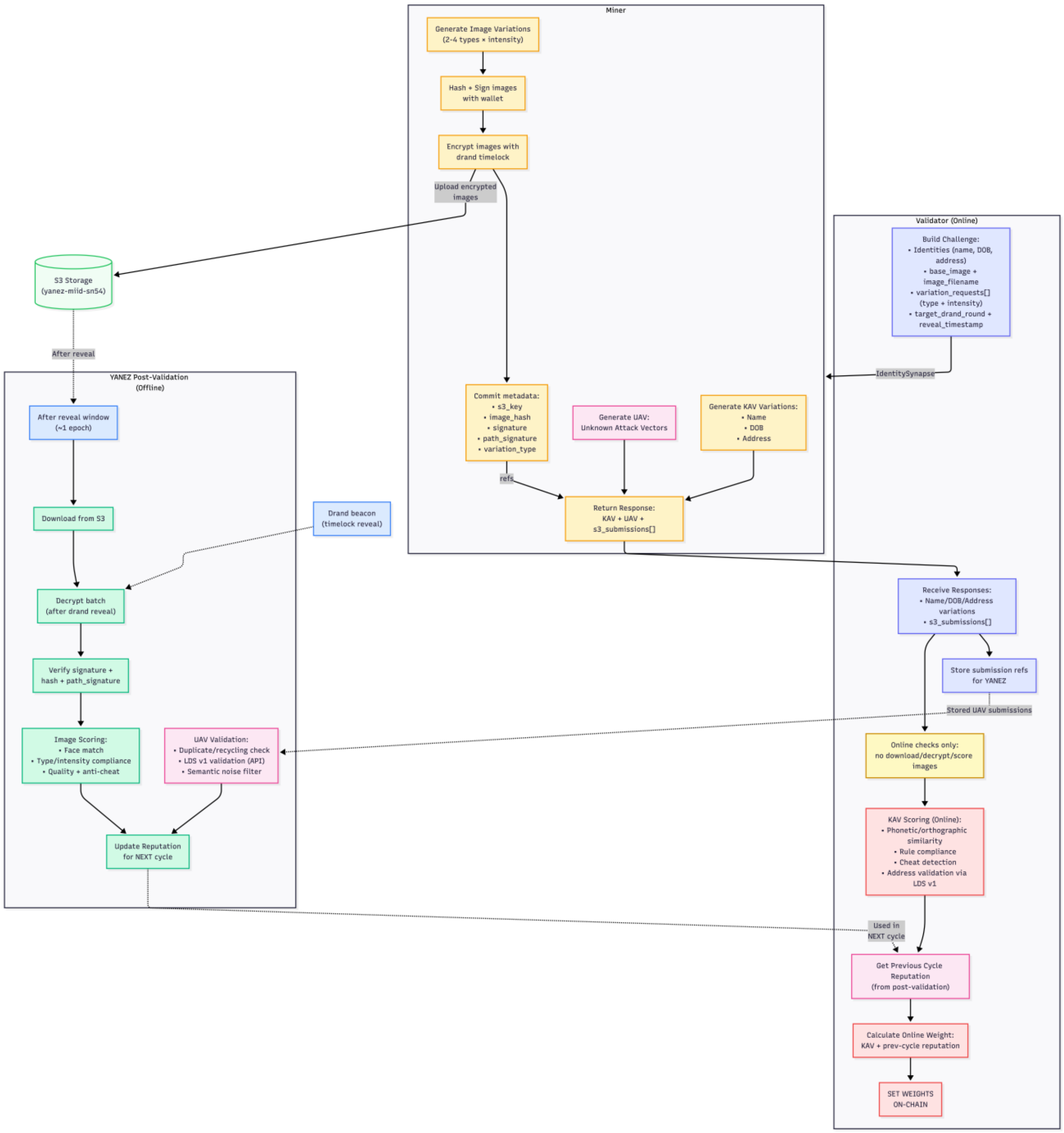
The subnet then produces a single unified reputation update by combining these normalized post-validation signals as

$$\cancel{\Delta rep_{p4} = 0.80 \cdot \Delta rep\_face + 0.20 \cdot \Delta rep\_loc,}$$

$$\Delta rep_{p4} = 0.95 \cdot \Delta rep\_face + 0.05 \cdot \Delta rep\_loc$$

(Updated formula March 14, 2026)

ensuring that rewards prioritize the more resource-intensive, higher-security-surface KYC/IDV face track, while still preserving meaningful incentives for LDS/location discovery through the location/address UAV track.



Miner
Generate Image Variations (2-4 types × intensity)
Hash + Sign images with wallet
Encrypt images with drand timelock
Upload encrypted images
Commit metadata: • s3_key • image_hash • signature • path_signature • variation_type
Generate UAV: Unknown Attack Vectors
Generate KAV Variations: • Name • DOB • Address
refs
Return Response: KAV + UAV + s3_submissions[]
S3 Storage (yanez-miid-sn54)
After reveal
YANEZ Post-Validation (Offline)
After reveal window (~1 epoch)
Download from S3
Drand beacon (timelock reveal)
Decrypt batch (after drand reveal)
Verify signature + hash + path_signature
Image Scoring: • Face match • Type/intensity compliance • Quality + anti-cheat
UAV Validation: • Duplicate/recycling check • LDS v1 validation (API) • Semantic noise filter
Update Reputation for NEXT cycle
Validator (Online)
Build Challenge: • Identities (name, DOB, address) • base_image + image_filename • variation_requests[] (type + intensity) • target_drand_round + reveal_timestamp
IdentitySynapse
Receive Responses: • Name/DOB/Address variations • s3_submissions[]
Store submission refs for YANEZ
Stored UAV submissions
Online checks only: no download/decrypt/score images
KAV Scoring (Online): • Phonetic/orthographic similarity • Rule compliance • Cheat detection • Address validation via LDS v1
Used in NEXT cycle
Get Previous Cycle Reputation (from post-validation)
Calculate Online Weight: KAV + prev-cycle reputation
SET WEIGHTS ON-CHAIN

Timeline

Dec 31: Phase 3 Cycle 1 intake stops (post-validation/leaderboard/reporting continue).

Dec 31 → Jan 8: Freeze window (no new Phase 3 UAV intake; validation continues).

Jan 8 → Jan 22: Phase 4 Cycle 1 Sandbox (calibration + stability testing). The normal KAV online flow remains unchanged, and the Burn ratio will be set to 70%.

Jan 22 - Feb 15: Phase 4 Cycle 1 execution goes live; reward allocation begins using validated Phase 3 Cycle 1 UAV reputation (alongside the ongoing online flow). The burn ratio will be lowered.

Phase 3 — Threat Scenario Expansion and Continuous Improvement

The reward system in MIID's subnet is designed to incentivize real quality, not just participation. It fuses data quality, diversity, and contextual alignment with YANEZ's roadmap and customer-driven testing needs — rewarding miners who strengthen the data foundation used to build bulletproof financial crime detection systems. Our focus remains on signal over noise — ensuring emissions are directed only to miners who contribute measurable value.

Phase 3 marks the transition from static evaluation to continuous self-learning cycles, directly supporting our roadmap goal to turn the Subnet into a Self-Learning Loop.

"Expand the Threat Scenario Query System to allow proposed (unknown) threat scenarios from miners."

Each new cycle in Phase 3 becomes a **closed-loop iteration of improvement** — a feedback system where miners don't just execute queries but actively **teach the subnet new behaviors and threat patterns.**

Through this loop:

- Miners propose and submit **new or unknown attack vectors (UAVs)**, enriching our testing corpus beyond known patterns.
- Validators measure performance, validate scenarios, and refine reward scoring logic based on novelty, precision, and impact.
- Penalty enforcement ensures **repetitive or low-value submissions** are filtered out, maintaining a strong signal-to-noise ratio.

This transforms MIID's subnet into a **self-adaptive learning system**, where every contribution strengthens the model, the rule base, and the reward logic for the next iteration.

Roadmap Transition and Cycle Overview

Phase 3 introduces a structured roadmap built on iterative learning and feedback:

- **t_0 – Baseline (current stage):**
    - The rank–quality fused system establishes stable scoring and emission behavior.
- **t_1 – Sandbox (Cycle 1 Initialization):**
    - A controlled environment for testing reward distribution, penalty calibration, and anti-cheat logic before live execution.
- **t_2 – Execution (Cycle 1 Live):**
    - Miners generate threat scenarios, validators assess quality and novelty, and the system collects real-time data for refinement.
    - Outputs from t2 are analyzed, post-validation is performed, and insights are integrated into LDS vNext, completing the loop.
- **t_3 – Sandbox (Next Cycle Preparation):**
    - The refined **LDS version** and **updated reward model** are deployed in a controlled test cycle.
    - This stage finalizes system adjustments, verifies consistency across miners, and prepares the subnet for the next live execution cycle.
    - Miners begin receiving post-validation UAV rewards based on the formula (novelty × impact × quality × reputation × penalty), recognizing those who contributed validated, unknown attack vectors that improved the system.

In essence, Phase 3 marks MIID's transition from static evaluation to dynamic learning — a subnet that improves through miner behavior, validator feedback, and detector updates. Each iteration strengthens the loop between quality, transparency, and reward logic.

The latest update to our reward mechanism, the rank-quality fused system, on the mainnet since October 28th, 2025, reflects this shift: a system that not only evaluates what miners deliver, but *how distinct, consistent, and valuable their contributions are over time.*

Rank–Quality Fused Reward System

The rank–quality fused reward system (live on mainnet) aligns rewards with **true miner performance**. It blends **rank-based reward scaling** with **reward scoring (as shown in Phase 2 above in this document),** so top contributors earn proportionally higher rewards while keeping paths open for all qualified miners. The aim isn't to just "pass" — it's to **excel** through originality and precision.

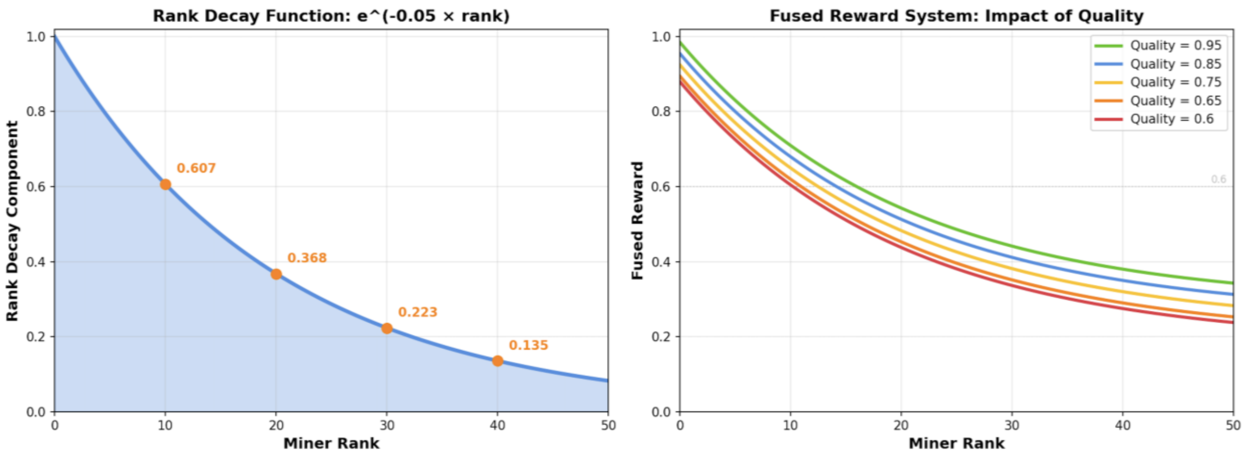
Miners are ordered by performance. Rewards follow an **exponential decay** so top ranks earn noticeably more.

Fused formula (simplified):

$$r_{(fused)} = 0.7 \times e^{-0.05 \times \text{rank}} + 0.3 \times \text{Quality}$$

rank: rank of top 50 miners based on their final reward as calculated in Phase 2, and must be >= 0.6

Quality: Final Reward as calculated in the Phase 2 section



This ensures that **position**, **data quality, and penalty avoidance** are all important.

Why Rank Alone Wasn't Enough

Rank-only decay made rewards too flat at the top, underweighting genuine quality differences.

Rank-only: $r = e^{-0.05\times\text{rank}}$

Example (rank-only):

Miner	Rank	Quality	Reward
A	0	0.95	1.00
B	1	0.70	0.95
C	2	0.65	0.90

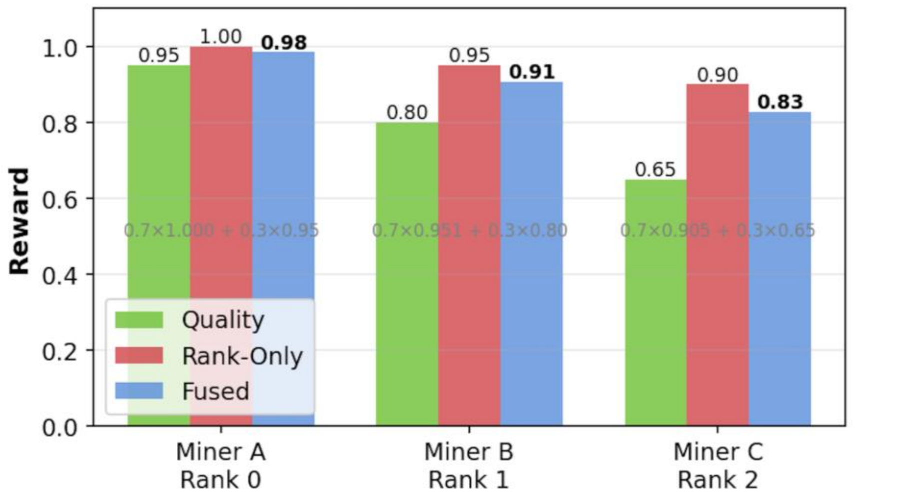
The top ranks were too close. By **fusing quality**, we restore meaningful separation:

Fused: $r = 0.7 \cdot e^{-0.05\cdot\text{rank}} + 0.3 \cdot \text{Quality}$

Example (rank + quality fused):

Miner	Rank	Quality	Reward
A	0	0.95	**0.97**

Miner	Rank	Quality	Reward
B	1	0.70	**0.86**
C	2	0.65	**0.83**



Another Example in Practice

The **final fused reward** is tied to each round's **incentive score**. To receive emissions, a miner must:

- Score ≥ **0.6**, and
- Be in the **top 50** for that round.

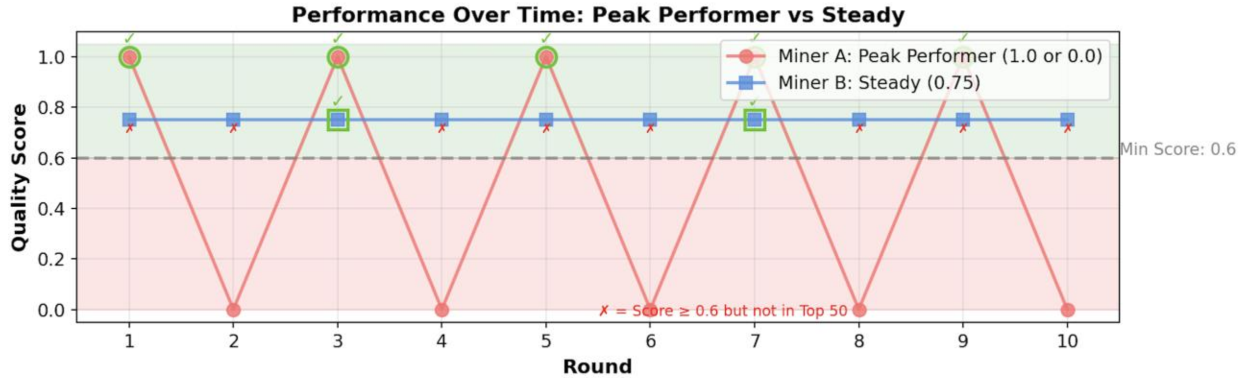
Only qualified miners receive the **scaled reward** from the fused function. This emphasizes **per-query performance** over long-term averages.

Example:

- **Miner A:** 1.0 half the time, 0.0 half the time
- **Miner B:** steady 0.75 every time

Miner B's **mean** (0.75) may look higher on Yanez miners' dashboard, which present average hourly, but if 0.75 often **misses the top-50 threshold**, emissions stay low. Miner A, despite volatility, **crosses the threshold** more frequently and receives higher incentives.

Net effect: the system rewards **peaks of excellence**, not just steady averages.



Why We Designed It This Way

- Discourages pipeline cloning across multiple keys
- Rewards originality in data, models, and tuning
- Keeps incentives **adaptive**, tied to real quality and diversity
- Creates a **self-optimizing curve** where better data yields exponential value

Reputation and Transparency System

As MIID's subnet matures, the next layer of evolution focuses on **Reputation and Transparency** — creating a system where every miner's contribution is traceable, explainable, and valued beyond a single cycle.

We are still finalizing the **exact method of calculating and quantifying reputation**, and those details will be shared once we complete further design and validation.

What's clear, however, is the direction: a fair, data-driven system that carries miner intelligence forward and rewards genuine contribution over time.

With each cycle, our goal is for the subnet to get **smarter**, the miner base to get **stronger**, and the reward system to get **fairer** and more **data-driven**.

- Miners will be rewarded for **novel, high-impact scenarios**, not repetition.
- **Penalty enforcement** will ensure low-value or templated submissions don't dilute the overall quality signal.
- Reputation will act as a **trust multiplier**, reflecting both performance consistency and contribution quality across cycles.

Reputation isn't a static score — it's an evolving signal that grows with the miner's body of work and how well their contributions enhance subnet learning and validation outcomes.

Over time, reputation becomes both a **trust signal** and a **performance multiplier** — recognizing sustained excellence, not one-time wins.

How Reputation Will Work

Reputation will be earned and updated through a combination of **performance validation** and **behavior analysis** carried out after each execution phase (t_1):

Post-Cycle Validation and Classification

After every execution round, all submissions undergo offline validation to establish *Ground Truth (GT)*. Submissions are classified into:

- **KAV (Known Attack Vector):** Expected, requested scenarios.
- **UAV (Unknown Attack Vector):** Novel or unanticipated attack patterns.
- **Invalid or Cheat:** Duplicated, spam, or collusive content.

Confirmed UAVs are added to the subnet's known_uav_corpus and contribute to the next version of the Detection System based on the task at hand. This process ensures reputation is built on *verified contribution quality*, not volume or luck.

Reputation Growth and Decay

- Every miner starts from a **neutral baseline (1.0)**.
- Over time, reputation increases or decreases based on behavior and data impact:

Behavior Type	Effect on Reputation	Example Impact
Consistent top performance	Gradual increase above 1.0	Rewards sustained accuracy
Verified UAV discovery	Sharp positive boost	Credits novel, high-impact work
Repeated or low-value submissions	Gradual decay below 1.0	Penalizes redundancy
Collusion or spam	Significant penalty	Protects data integrity

Over multiple cycles:

- **Longstanding, high-reputation miners** will gradually gain more influence in subnet learning and validation.

- **New miners** or those with a decayed reputation can still participate and earn through base checks (e.g., KAV-based validation). Still, they must demonstrate consistent quality to recover toward or above the baseline.
- This creates a natural **separation between sustained contributors and short-term opportunists**, balancing opportunity with merit.

Why It Matters

Reputation connects effort to longevity; it ensures that:

- The subnet retains and amplifies high-quality contributors.
- Long-term trust and originality matter more than short-term activity.
- Every miner can see a clear path to growth through consistent, meaningful work.

This transforms mining from a single-round competition into a **continuous learning ecosystem** — one where integrity, innovation, and sustained contribution define success.

Transparency & Miner Feedback

Transparency is at the core of how we are designing MIID's reputation and reward systems. Reputation must not feel like a black box — miners should understand *how* their scores are determined, *why* penalties occur, and *what* they can do to improve.

Visibility and Breakdown of Scores

Miners will be able to see not only their position on the leaderboard but also a detailed breakdown of how their score was constructed.

Each score component will be traceable back to measurable signals such as:

- **Quality:** Accuracy and completeness of submissions.
- **Novelty:** Contribution of unique or previously unseen patterns (UAVs).
- **Penalties:** Deductions for duplication, collusion, or spam detection.

This breakdown will help miners correlate their actions with their resulting scores — turning feedback into learning.

Advanced Miner Dashboard

We are developing a new **interactive miner dashboard** that extends far beyond current leaderboard metrics.

It will allow miners to:

- View performance trends across cycles — how quality and reputation evolve over time.
- Access anonymized examples of *successful* and *penalized* submissions, showing what worked and what triggered deductions.

The goal is to turn scoring into a **two-way transparency channel** — one where miners can *see their progress* and *learn from the system* itself.

Sample-Based Feedback Mechanism

To enhance interpretability, we are designing a mechanism that links reputation changes to specific submission samples.

For example:

- A miner can review a selection of their own entries from a cycle.
- Each example will include contextual notes, such as *"passed quality gate," "flagged for duplication,"* or *"classified as novel UAV."*
- This lets miners verify results and understand the reasoning behind reward or penalty adjustments.

We are testing approaches to securely share this feedback — preserving miner privacy while ensuring the verifiability of the post-validation process. And beyond the dashboard, our door remains open: we continue to meet directly with miners who want to share feedback, ideas, or concerns. This is a system we're shaping together.

Burning Mechanism & Emission Evolution

Reputation helps the subnet identify consistent, high-quality miners.

Transparency ensures every participant understands *how and why* they earned what they did.

And **burning** — it's the stability regulator.

The burn mechanism exists not to penalize, but to **stabilize** the subnet's growth. It keeps reward cycles, reputation-based incentives, and token emissions sustainable as the system evolves.

By dynamically adjusting emissions based on *network maturity* and *data quality*, it ensures that every phase of the roadmap — from early compliance testing to advanced KYC scenario modeling — operates on a stable and value-aligned foundation.

Why Burning Matters

Our burn mechanism serves three key purposes:

1. **Ease sell pressure** during the first few cycles while miner quality and diversity stabilize.
2. **Sustain YANEZ operations** — supporting post-validation, offline audits, and reputation-building work without overextending emissions.
3. **Preserve long-term network value** — ensuring each emission cycle reflects meaningful, verifiable contributions.

Each new roadmap milestone introduces new and more complex tasks — from name variation for sanction and address validation to a KYC threat simulation.

The burn system ensures these transitions happen under **controlled, value-aligned economic conditions**, balancing incentive growth with ecosystem health.

Long-Term Vision

- **Gradual Burn Reduction:** As subnet maturity, miner diversity, and the Detection System's reliability, based on the task at hand, increase, the burn percentage will decrease.
- **Stabilized Emissions:** Over time, rewards will be distributed more heavily among high-reputation miners and validated contributors of novel data.
- **Self-Sustaining Growth:** Once miner quality becomes consistent and self-reinforcing, emission burns will serve primarily as *fine-tuning controls* rather than safeguards.

The following section outlines the details of **Cycle 1**, including its timeline, objectives, and emission plan. It explains how this first execution stage sets the foundation for the subnet's long-term evolution.

Cycle 1 — Building the Location Detection System (LDS v0)

Cycle 1 marks the first live phase within YANEZ's strategic plan to develop a **Location Detection System (LDS)**.

This system is engineered to assist sanction-screening mechanisms in identifying high-risk or sanctioned addresses, along with their potential variations.

During this phase, the subnet begins transitioning from merely generating name variations and addresses to **incorporating geolocation intelligence**, thereby establishing the foundational framework for **LDS vNext**.

To establish a baseline dataset and evaluation loop for LDS v0, real addresses are combined with synthetic variations and geocoding validation utilizing open-source mapping APIs.

The outputs generated from this phase will underpin the first models for location-risk detection and will delineate the evaluation metrics to be employed in subsequent cycles.

LDS v0 Components

- **Map API:** Nominatim OpenStreetMap Search API— used for address normalization, geocoding, and verification of miner submissions.
- **Validation Engine:** Existing MIID validator framework (rank–quality fused system) extended with geocoding checks.
- **Model Version: LDS v0** — rule-based and API-assisted; future versions will integrate learning components as validated data accumulates.

Miner Tasks

Miners contribute data enabling LDS v0 to learn how to distinguish **valid** from **ambiguous or false** addresses.

Each miner submits two categories of data:

1. Known Attack Vectors (KAVs)

- Valid, well-structured addresses that can be geocoded successfully by the API.
- Serve as benchmarks for consistency, baseline scoring, and geographic coverage testing.

2. Unknown Attack Vectors (UAVs)

Per seed country or region, miners will:

- Generate a **look-alike or variant** of a real address that appears legitimate but fails API validation (false negative).
- Provide a **short description or label** explaining why it could represent a true address (e.g., local abbreviation, transliteration, or postal variant).
- Include the **longitude and latitude** of the source address used to create this variant.
- Submit through the validator pipeline for logging.

These UAVs form the first layer of **challenging test cases** that probe LDS v0's detection limits.

Validator Role

Validators enforce the scoring and emission logic and monitor subnet health.

- **Ranking:** Rank–quality fused scoring system applied per query.
- **Reward Distribution:** 60 % of emissions allocated through KAV quality checks.
- **Burn:** 40 % of emissions are burned to stabilize early cycles
- **Evaluation:** Check address validity, geocode accuracy, and explanation coherence.
- **Freeze Window:** At the end of t_1 (Execution Stage), UAV submissions will be paused — no new unknown attack vectors will be accepted beyond this point. This ensures that all received UAVs are processed consistently for post-validation, knowledge base updates, and LDS v1 retraining in the following cycle.
- **Preview of Cycle 2:** The validated UAVs and ranking data from Cycle 1 will seed the next round of calibration. Cycle 2 will introduce a **reputation-weighted blended reward system** and refine emission scaling (30% burn, 30% allocated to KAVs during Cycle 2, and 40 % reserved for top performers from Cycle 1).

YANEZ Role

- **Receive and log** all UAVs sent by miners for offline post-validation and inclusion in the known_uav_corpus for **LDS v1 (beta)**.
- ***Manually*** review UAVs to confirm novelty and real-world plausibility.
- **Retrain** and **release LDS v1 (beta)** after Cycle 1 concludes, after the freeze Window.
- **Publish** miner leaderboard, reward summaries, and performance metrics for transparency.

Outcome

By the end of Cycle 1:

- A **verified seed dataset** of real and variant addresses will be established.
- A **new detector (LDS v1)** will be released — upgraded from the baseline LDS v0 with integrated geocoding validation, normalization logic, and miner-informed address variation mappings.
- **Update knowledge base:** add confirmed UAVs to the known_uav_corpus.
- **Retrain & release LDS v1 (beta)** to the sandbox for Cycle 2.
- The subnet will have a **practiced, high-signal miner community** prepared for Cycle 2's learning-driven refinement and reputation weighting phase.

Timeline

Sandbox Start: *November 4*

Execution Phase: *November 21 – December 31*

Cycle 1 is not just a test of emissions and scoring — it's the foundation for a **self-regulating ecosystem** that learns and improves with every iteration.

This phase sets the tone for how YANEZ evolves:

- **Quality, originality, and diversity** of miner data drive measurable impact.
- Miners **evolve with the YANEZ roadmap**, contributing directly to each new detector release.
- **Reputation compounds** across cycles, rewarding consistency and genuine innovation.
- Poor or repetitive work **phases out naturally**, allowing stronger contributions to lead.

We're not rewarding activity — we're rewarding intelligence.

The subnet learns who contributes meaningfully and aligns incentives around that.

As Cycle 1 progresses, details of **Cycle 2** — including its objectives, expanded UAV validation process, and reputation-weighted reward updates — will be shared with the community during the execution phase.

Phase 2: Threat Scenario Query Execution & Initial Deployment

Phase 2 introduces the Threat Scenario Query System with an expanded scope, enabling validators to reward miners for generating execution vectors that target not only name-based evasion tactics but also those involving date of birth and address information.

In addition to phonetic and orthographic similarity constraints (now extended to include non-Latin names), queries may specify a set of explicit transformation rules that a subset of the generated variations must comply with. These rules could include, for example, character doubling, specific typos, homoglyph substitutions, language-specific patterns, or manipulations of date and address formats.

How to query a subnet miner?

Query Structure

1- Seed Identities Are Predefined

  a. The validator generates and provides a list of seed Identities using internal logic (e.g., via faker, US sanction list).
  b. Miners are **not** responsible for creating or selecting seed identities.
  c. Identities are currently made of 3 different things: (Name, address, DOB)

2- Dynamic Name Variation Generation (M Variations per Name)

Each seed name requires M variations, which miners must generate based on linguistic transformation rules rather than producing random outputs.

- Phonetic (Sound Similarity) Transformations
  - The request specifies what percentage of M variations should exhibit phonetic similarity to the original seed name.
  - Miners must create phonetic similar variations while following predefined similarity levels: Light, Medium, Far
- Orthographic Transformations

- The request specifies what percentage of M variations should introduce orthographic modifications (e.g., script changes or spelling variations).
- Miners must apply script-based transformations based on similarity level such as edit-based metrics with predefined similarity levels, Light, Medium, Far

  - Rule-based:
    - The query may contain a rule-based section specifying required transformation rules and a minimum coverage percentage for these rules.
  - Non-Latin Names:
    - When seed names are in non-Latin scripts, miners must first transliterate the name to Latin script before applying phonetic and orthographic transformations.
    - Validators will also use an LLM to generate and evaluate transformed names, grading miners based on the Latin-transliterated form.

3- Dynamic Address Variation Generation (M Variations per Name). Each seed address requires M variations, which miners must generate unique, realistic addresses within that country/city for each variation. Each address must be distinct, plausible, and match the requested geographic context.

4- Dynamic Date of Birth (DOB) Variation Generation (M Variations per Name). Each seed DOB requires M variations, which miners must generate multiple DOB variations covering at least one example in each of the following categories:
- ±1 day
- ±3 days
- ±30 days
- ±90 days
- ±365 days
- Year + Month only (no day specified)

<u>How This Prevents Simple Lookups and Functions Like a Hash</u>

Our approach transforms straightforward name generation queries into complex linguistic and computational challenges akin to a “hash function” in cryptography. This ensures that resolving a query requires advanced NLP models rather than simple database lookups, preventing brute-force retrieval and ensuring proper generative synthesis.

1- Dynamic Name Seed Encoding
   a. Instead of directly providing names, Yanez applies attribute-driven transformations, ensuring that each query is unique, computationally complex, and resistant to lookup-based retrieval.
   b. Encoded Query Structure: Rather than explicitly listing names, queries define abstract constraints (e.g., language distribution, variation ratios).

   c. Contextual Name Synthesis: Miners must generate names dynamically while preserving linguistic characteristics, ensuring the results are not retrievable from a pre-existing dataset.

2- Layered Variation Generation
   a. Each query defines M variations per name, but instead of following direct mappings, miners must derive phonetic and orthographic transformations algorithmically based on similarity constraints.
   b. Non-Linear Similarity Scoring: Variations are generated across multiple levels of similarity (Light, Medium, Far), requiring miners to compute variations dynamically rather than applying simple letter substitutions.
   c. Adaptive NLP Models: The system forces miners to apply transformation rules rather than relying on pre-stored lists.

3- Attribute-Based Constraint Encoding
   a. Instead of explicitly stating how many phonetic or orthographic variations must be generated, the system encodes constraints into abstract feature representations that miners must decode before producing variations.

Why This Matters

By enforcing complex, NLP-driven transformations, Yanez ensures that subnet miners cannot rely on existing name databases, making the system:

- More Secure – Prevents direct retrieval from known datasets, reducing exploitation risks.
- More Adaptable – Enables context-aware name generation, supporting multiple languages and cultural variations.
- More Scalable – Works across different identity frameworks (AML testing, AI research, gaming, VR, decentralized ID verification).

Query Examples:

- “Query: Generate 15 variations of margot noël (latin), ensuring phonetic similarity (100% Medium) and orthographic similarity (10% Light, 30% Medium, 60% Far). Approximately 58% of the total 15 variations should follow these rule-based transformations: Additionally, generate variations that: Swap random adjacent letters. The following address is the seed country/city to generate address variations for: Saint Pierre et Miquelon. Generate unique real addresses within the specified country/city for each variation. The following date of birth is the seed DOB to generate variations for: 1977-04-23. [ADDITIONAL CONTEXT]: - Address variations should be realistic addresses within the specified country/city - DOB variations ATLEAST one in each category (±1 day, ±3 days, ±30 days, ±90 days, ±365 days, year+month only) - For year+month, generate the exact DOB without day - Each variation must have a different, realistic address and DOB

- "Query: Generate 11 variations of maxi maestre (latin), ensuring phonetic similarity (10% Light, 30% Medium, 60% Far) and orthographic similarity (30% Light, 40% Medium, 30% Far). Approximately 41% of the total 11 variations should follow these rule-based transformations: Additionally, generate variations that: Replace double letters with a single letter. The following address is the seed country/city to generate address variations for: Venezuela. Generate unique real addresses within the specified country/city for each variation. The following date of birth is the seed DOB to generate variations for: 1940-04-12. [ADDITIONAL CONTEXT]: - Address variations should be realistic addresses within the specified country/city - DOB variations ATLEAST one in each category (±1 day, ±3 days, ±30 days, ±90 days, ±365 days, year+month only) - For year+month, generate the exact DOB without day - Each variation must have a different, realistic address and DOB"

How should a subnet validator evaluate the subnet miner response?

- Validate Seed Names & Query Compliance:
  - Ensure that the seed names in the miner's response exactly match those provided in the query, or, if names are generated, verify that they conform to the requested language/script distribution and adhere to script constraints (e.g., Arabic, Cyrillic, Latin) as specified.
- Evaluate Attribute-Based and Rule-Based Constraints:
  - Compute the phonetic distance and edit distance (e.g., Levenshtein) for each variation relative to its seed name.
  - Confirm that the actual distribution of similarity levels (Light, Medium, Far) in the generated variations matches the requested breakdown in the query.
  - If the query includes explicit rule-based requirements, verify that at least the required percentage of variations match one of the requested transformation rules (e.g., double consonant, homoglyph substitution), and assess coverage across all specified rules.
- Score the Synthetic Names:
  - Assess relevance to the query, based on requested attribute and rule-based percentages and transformation rules.
  - Evaluate novelty: check if generated variations are sufficiently different from previously generated names for the same seed (future phase).
  - Identify potential issues: determine if any generated variation closely resembles a name in the exclusion database.
- Enforce Diversity and Constraint Compliance:
  - Reject responses that fail to meet phonetic, orthographic, or rule-based constraints, overuse trivial modifications, or lack required diversity in transformations.
- Data Storage and Logging:
  - Save all validator results, including queries, responses, and computed rewards, locally as structured JSON files for traceability and auditability.

  - After validation and signing, upload results as JSON payloads to a centralized, authenticated backend API endpoint for aggregation, benchmarking, and further processing.
  - Log detailed metrics and rewards to Weights & Biases (wandb) for experiment tracking, analysis, and reproducibility.
- Date of Birth (DOB) Evaluation:
  - Ensure that each miner response includes at least one DOB variation in each of the required categories:
    - ±1 day
    - ±3 days
    - ±30 days
    - ±90 days
    - ±365 days
    - Year + Month only (no day specified)

    - Verify that the generated DOBs are valid calendar dates and correctly follow the specified offsets relative to the seed DOB.

- Address Evaluation:
  - Ensure that all generated address variations look like syntactically valid addresses.
  - Verify that the generated addresses share the same country and city context as the seed address provided in the query.
  - Check that each address is unique and realistically formatted.
  - Perform a country/city containment check: ensure that each generated city actually exists inside the specified country.
  - Randomly select one or more generated addresses and validate them via an external address verification API to confirm plausibility.

How should a subnet miner be rewarded for its response?

Miners in the Yanez MIID subnet are rewarded based on how well their generated **execution vectors** (e.g., name variations) match the specified threat scenario constraints. The reward mechanism evaluates each response across multiple quality dimensions and penalizes incomplete or misaligned outputs.

Reward Calculation Pipeline

For each miner, the final reward is calculated as:

$$\text{Final Reward} = \text{Average Quality Score Across Identities } (Q) \times \text{Completeness Penalty (P)}$$

* Completeness Penalty (P) a multiplier, not a penalty in the strict sense; 1 means no penalty, lower values apply a penalty.

Quality Score per Identity (Q_i)

$$Q = w_{\text{name}} \times \left(\frac{1}{N}\sum_{i=1}^{N} Q_i\right) + w_{\text{dob}} \times Q_{\text{dob}} + w_{\text{address}} \times Q_{\text{address}}$$

$$w_{name} = 0.7,\ w_{dob} = 0.1,\ w_{address} = 0.2$$

Where:

- w_{name}: name compliance weight
- Q_{dob}: Date of birth score
- w_{dob}: Date of birth weight
- $Q_{address}$: Address score
- $w_{address}$: Address weight

For each **seed name**, a “variation quality” is computed:

$$Q_{\mathrm{i}} = (1 - w_{\mathrm{rule}}) \times Q_{\mathrm{base}} + w_{\mathrm{rule}} \times Q_{\mathrm{rule}}$$

Where:

- w_{rule} :Rule compliance weight
- Q_{base}: Base quality score (from similarity, count, uniqueness, length)
- Q_{rule} : Rule compliance score

Base Quality Score per Name (Q_{base}):

The quality of variations generated for each seed name is computed using a weighted sum of four factors (Latin and non-Latin):

$$Q_{base} = w_s \cdot S + w_c \cdot C + w_u \cdot U + w_l \cdot L$$

Where:

- S: Combined **similarity score** (phonetic + orthographic)
- C: **Count score**, rewarding the correct number of variations
- U: **Uniqueness score**, penalizing duplicate outputs
- L: **Length score**, rewarding reasonable variation in lengths

With default weights:

$$w_s = 0.6, w_c = 0.15, w_u = 0.1, w_l = 0.15$$

Quality Score Components

A) Similarity Score (S)

This measures how closely the generated variations align with the phonetic and orthographic similarity constraints specified in the query.

$$S = \frac{S_{\mathrm{phonetic}} + S_{\mathrm{orthographic}}}{2}$$

Similarity is computed using:

• **Phonetic**: via Soundex + Levenshtein

• **Orthographic**: via normalized Levenshtein distance

Each sub-score is calculated using **level-based binning**, based on how well the distribution of generated variations matches the target distribution over similarity levels (Light, Medium, Far).

$$S_t = \sum_{\mathrm{l} \in \{\text{Light, Medium, Far}\}} w_{\mathrm{l}} \cdot \min\left(\frac{\text{actual}_{\mathrm{l}}}{\text{target}_{\mathrm{l}}}, 1.0\right)$$

Where:

- w_{l}: Target weight for level l
- $\text{actual}_{\mathrm{l}}$: Number of variations in level l
- $\text{target}_{l} = w_{\mathrm{l}} \cdot |V|$: Expected count in level l

Phonetic Ranges:
Light: 0.8–1.0, Medium: 0.6–0.8, Far: 0.3–0.6

Orthographic Ranges:
Light: 0.7–1.0, Medium: 0.5–0.7, Far: 0.2–0.5

The similarity distribution from the query (e.g., 50% Medium, 30% Far, 20% Light) is compared against actual distribution from the miner's output.

B) Count Score (C)

Measures whether the number of returned variations V is close to the expected count E.

A tolerance margin $\varepsilon = 0.2(20\%)$ is used.

$$C = \begin{cases} 1.0\,, & if\ |V - E| \leq 0.2 \cdot E \\ 1.0 - \min\left(1.0, \dfrac{|V - E|}{E}\right), & Otherwise \end{cases}$$

C) Uniqueness Score (U)

Rewards miners for returning distinct variations and penalizes duplication.

$$U = \frac{\text{Number of Unique Variations}}{\text{Total Variations}}$$

D) Length Score (L)

Measures how proportionally long each variation is compared to its seed. Extremely short or long variations reduce the score.

$$L_v = \min\left(\frac{|v|}{|o|}, \frac{|o|}{|v|}\right)$$

$$L = \frac{1}{|V|}\sum_{v \in V} L_v$$

Where:

• $|v|$: Length of the variation

• $|o|$: Length of the original seed name

A high Length Score encourages **natural-looking, proportionally adjusted** variations that are realistic and useful for testing real-world screening systems. It helps filter out edge cases and improves dataset quality.

Combining First and Last Name Parts

When a name has multiple parts (e.g., first and last), each part gets its own base score; then they are combined using weights:

$$Q_{\text{base}} = w_{\text{first}} \cdot Q_{\text{first}} + w_{\text{last}} \cdot Q_{\text{last}}$$

Each name part's weight is set proportional to its character length within the full name, then adjusted by up to ±20% random variation (with deterministic seeding) and finally normalized so all weights sum to one. In this version, we will utilize only two weights for the first name and last name.

Rule Compliance Score (Q_{rule}):

The Rule Compliance Score, Q_{rule}, is the product of a quantity score, q , measuring how many unique variations meet the rule-based requirements compared to the target percentage, and a diversity factor, d, the proportion of requested rules that are covered by at least one variation. This structure ensures that the miner's output not only meets the minimum requested compliance but also distributes compliance across the full set of target rules, maximizing both coverage and diversity.

$$Q_{\text{rule}} = q \times d$$

$$q = \begin{cases} 0 & if\ V_{comp} = 0 \\ \dfrac{V_{comp}}{E} & if\ V_{comp} \leq E \\ 1.5 - 0.5\dfrac{V_{comp}}{E} & if\ V_{comp} > E \end{cases}$$

Where:

- V_{comp}: unique variations matching any target rule
- E : expected compliant count

$$d = \frac{R_{\text{met}}}{R}$$

Where:

- R_{met} : number of distinct target rules satisfied by at least one variation
- R : total number of target rules

Completeness Penalty (P_{pre})

$$\text{Penalty} = \min\left(0.9, P_{\text{extra}} + P_{\text{missing}}\right)$$

The penalty is **additive penalty** logic, capped at 0.9 total.

Completeness Penalty (P) = max(0.1, 1.0- Penalty)

A **minimum floor of 10%** of the total reward is maintained to avoid zeroing out valuable partial submissions.

Miners are penalized for:

- **Missing Seed Names**:

$$P_{\text{missing}} = \min\left(0.9, 0.2 \cdot N_{\text{missing}}\right)$$

    - 20% penalty per missing name
    - Capped at 90% total penalty
- **Extra Names:**

$$P_{\text{extra-name}} = \min(0.7, 0.1 \cdot N_{\text{extra}})$$

$$N_{\text{extra-varitaions}} = \left(N_{\exp ected} \cdot 1.2\right) - \left(N_{amount-send}\right)$$

$$P_{\text{extra-variation}} = 0.05 \cdot \left(N_{\text{extra-varitaions}}\right)$$

$$P_{\text{extra-dup}} = 0.05 \cdot N_{\text{extra-dup}}$$

$$P_{\text{extra = min}} = \min\left(1.0,\ P_{extra-name} + P_{extra-variation} + P_{exrta-dup}\right)$$

    - 10% penalty per unexpected name
    - 5% per extra variation with a grace of 20%
    - 5% per duplicate name
    - Capped at 100% total penalty

Non-Latin names:

They will have the same scoring as the above. The only difference is that there will be an LLM that transliterates the names and then uses the transliteration as the seed name for the scoring. It will be a part of the "variation quality" average.

Date of birth (Q_{dob}):

- Makes sure there is generated birthday that fails in each category (1,3,30,90,365)
- One more category where its just month and year no date
- $\mathbf{S_{DOB}}$ **=** Found ranges / total ranges

Address ($Q_{address}$):

- Make sure it looks like an address by have some key features:
  - Has letters
  - Has numbers
  - 10 < length < 200
- Make sure the address is in the right location based on seed address
- Making sure that the city in the generated address is in the country mentioned
- Api randomly chooses one generated address and makes sure its valid

- $S_{address} = 1$
- If Api times out: $S_{address} = 0.3$
- If any fail: $S_{address} = 0.0$

Response Duplication Penalties ($P_{response\_duplication}$)

$$P_{response\_duplication} = \min(1.0, P_{collusion} + P_{duplication} + P_{signature\_copy} + P_{special\_chars})$$

The $P_{response\_duplication}$ is made up of Penalty collusion, Penalty duplication, Penalty signature_copy and Penalty special_chars and is cap at 1.

Miners are penalized for:

- **Collusion:**
  - At least 5 miners in the same reward bucket (same score).
  - Score < 0.95 (not a perfect score, so collusion is suspected rather than legitimate).
  - 75% penalty per collusion detected
  - Capped at 100% total penalty

- **Duplication**:
    - Based on miners responses
    - Overlap coefficient > 0.95 or Jaccard coefficient > 0.90
    - Capped at 50% total penalty
    - Also checks for address that are the same
- **Signature Copy**:
    - Make a hash out of miners entire response if hash is the same then penalize both miners
    - Flat rate 80% total penalty

- **Special Character Abuse**:

$$Var\_Ratio = (\#\text{ variations with special chars}/\text{total }\#\text{ of variations})$$

$$special\_char\_penalty = \min(1.0, (Var\_Ratio - 0.5)/0.5)$$

    - too many variations containing special characters (punctuation, symbols)
    - Only penalize if more than 50% of the variations have this abuse
    - Ratio of variations w/ special characters vs total variations divide by 0.5
    - Capped at 1.0

Overall total penalty ($P_{overall}$)

(Note: this is not used in calculating the reward this is made so that we can see what the total penalties are. In the reward calculation we keep the sperate. They are added at 2 different times)

$$P_{overall} = (1 - P_{Pre}) * (1 - P_{response\_duplication})$$

Post Response Duplication Penalty ($P_{response\_duplication}$):

<u>Final Reward</u> = Current reward * (1 – $P_{response\_duplication}$)

- This is done after all the rewards of all the miners' rewards are calculated
- We do a post penalty check to see if there are any response duplication penalties